Roping in Uncertainty: Robustness and Regularization in Markov Games

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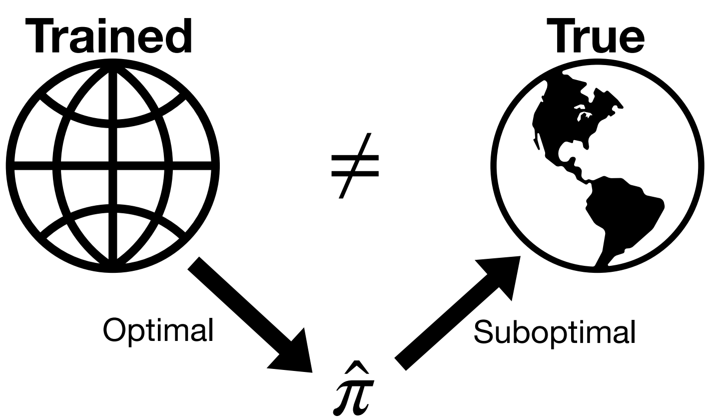
Robust Nash Equilibrium can be computed, sometimes *efficiently*, via Game Regularization!

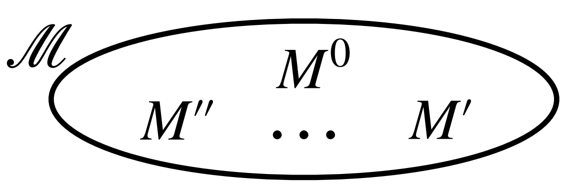
Motivation



Offline RL must be robust!

Robust Markov Games

RMG = model M^0 + uncertainty set $\mathscr{M}$:



Ex: $\mathscr{R}_{i,s,h} = \{R \in \mathbb{R}^{A_1 \times A_2} \mid ||R - R^0||_p \leq \alpha_{i,s,h}\}$

Robust NE (RNE)

$$\pi_i^* \in \arg\max_{\pi_i} \min_{M \in \mathscr{M}} V_{M,i}^{\pi_i^*, \pi_{-i}^*}$$

Regularized Games

RGs use a Regularized Value:

$$V_{i,h}^{\pi}(s) := E_M^{\pi}\left[\sum_{t=h}^{H} r_{i,t}(s_t, a_t) - \Omega_{i,t}(s_t, \pi_t) \mid s_h = s\right]$$

Example Regularizers:

1. Entropy: $\Omega_{i,h}(s, \pi) = \alpha_i \sum_{a_i \in \mathscr{A}_i} \pi_i(a_i) \log \pi_i(a_i)$
2. Norm: $\Omega_{i,h}(s, \pi) = \alpha_{i,s,h} ||\pi_i||_p ||\pi_{-i}||_q$

Equivalence of RMG and RG

Robust Markov Game

$\Updownarrow$

Characteristic function

$$\Omega_{i,h}(s, \pi) := \sigma_{\mathscr{R}_{i,s,h}}\left(-\pi_i \pi_{-i}^\top\right)$$

$\Updownarrow$ $\quad \sigma_X(y) := \sup_{x \in X} \langle x, y \rangle$

Regularized Game

*Similar results hold for Transition Uncertainty

Computational Complexity

$$-\sigma_{\mathscr{R}_1}(-\pi_1 \pi_2^\top) = \pi_1^\top R_1 \pi_2$$
$$-\sigma_{\mathscr{R}_2}(-\pi_1 \pi_2^\top) = \pi_1^\top R_2 \pi_2$$

$\Rightarrow$ General Sum

Theorem: *Computing RNE is PPAD-hard even for (s,a)-rectangular zero-sum matrix games*.

Key to Tractability: *Decomposability*

$$\sigma_{\mathscr{R}_{i,s,h}}(-\pi_i \pi_{-i}^\top) = \Omega_{i,i}^h(\pi_i) + \Omega_{i,-i}^h(\pi_{-i})$$

Theorem: *If σ is decomposable, the equivalent RG is zero-sum so can be solved in poly time*.

Conclusions

Robustness $\Leftrightarrow$ Regularization

- Robust policies can be found using regularization, efficiently for many uncertainty sets.
- Regularizers provide robustness.